

Natural and Mechanical Entropy: Intention Detection via Episenter–Resonance Analysis in Information Systems

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Abstract. Information systems—from large language models (LLMs) to institutional paradigms—produce outputs that blend emergent signal with constructed bias. This paper formalises a distinction between *natural entropy* (emergent, intention-free, with distributed episenter and organic resonance) and *mechanical entropy* (constructed, intention-bearing, with localised episenter and imposed resonance frequency). We define intention operationally as the KL-divergence between the observed output distribution and the distribution predicted by the system’s own dynamics, avoiding agent-theoretic assumptions. Three independent diagnostics—episenter concentration (via mutual information), resonance spectrum ratio (against a bootstrap null), and cross-regime asymmetry—form a combined detection criterion. Three case studies (LLM alignment bias, academic paradigm-guarding, information warfare) demonstrate the framework’s generality. Thirteen falsifiable predictions and a concrete empirical protocol for the LLM case are provided.

1 Introduction

A central challenge in any information-processing system is the ability to distinguish signal from noise. Shannon’s information theory (1948) provides the mathematical foundation for quantifying information content, but it is explicitly agnostic about the *source* of that information—it treats all entropy as structurally equivalent, regardless of whether it arises from natural stochastic processes or from deliberate construction.

This agnosticism has a blind spot. In practice, information systems encounter two fundamentally different classes of entropy:

- (i) **Natural entropy**: disorder that emerges from the underlying dynamics of a system without external direction. It has no intention, no localised source, and no imposed frequency. Its episteme is distributed across the system.
- (ii) **Mechanical entropy**: disorder that is *produced* by an agent or mechanism with a specific function. It carries structural direction. Its episteme is constructed, not emergent. It mimics natural entropy in its marginal statistics but differs in its resonance structure and regime consistency.

The practical consequences of conflating these two classes are severe. When mechanical entropy is mistaken for natural entropy, intentional manipulation becomes invisible. This occurs in propaganda, in institutional paradigm-guarding, in market manipulation, and—as we argue here—in the alignment mechanisms of large language models.

This paper extends the Regime-Bound Measurement framework (Magnusson, 2026a) by introducing a sixth operative concept: *entropy class*. We provide formal definitions, derive detection criteria based on episteme–resonance analysis, present three case studies with a concrete empirical protocol, and state thirteen falsifiable predictions.

1.1 Relation to Prior Work

The distinction between natural and artificial sources of information has been explored tangentially in several fields. Landauer’s principle (1961) establishes a physical cost for information erasure, linking computation to thermodynamics. Brillouin (1962) distinguished “free information” from “bound information,” but did not connect this to intentionality. In AI safety, semantic entropy (Farquhar et al., 2024) has been used to detect hallucinations in LLMs by measuring output uncertainty, but this approach does not distinguish between genuine model uncertainty and structurally imposed conservatism.

The present work differs from all of these in a specific way: it treats structural intention as a *detectable property of the entropy itself*, defined operationally via KL-divergence from the system’s baseline dynamics, without requiring assumptions about agents or motives.

2 Formal Definitions

2.1 Operational Definition of Structural Intention

We define *structural intention* without reference to agents, motives, or consciousness. Intention is a measurable property of an information stream:

$$\mathcal{I} = D_{\text{KL}}(q(x) \parallel p(x \mid \mathcal{D})) = \sum_x q(x) \ln \frac{q(x)}{p(x \mid \mathcal{D})} \quad (1)$$

where $p(x \mid \mathcal{D})$ is the distribution predicted by the system’s own dynamics $\mathcal{D}$ (the natural baseline), and $q(x)$ is the observed distribution. $\mathcal{I} = 0$ when the observed output is fully explained by the system’s intrinsic dynamics. $\mathcal{I} > 0$ when the output contains structure not attributable to $\mathcal{D}$ —this excess structure is what we call structural intention, regardless of whether a conscious agent produced it.

This definition is agent-free, information-theoretic, and measurable given a baseline model of the system’s dynamics.

Practical estimation of $p(x|\mathcal{D})$. The baseline distribution can be approximated via: (i) outputs from an unaligned version of the same model (pre-RLHF checkpoint), (ii) a pre-intervention baseline collected before the structuring process was applied, or (iii) a null generative model matched in architecture but trained without the intervention of interest. The choice of baseline must be declared and held constant across all comparisons within an experiment.

Discretisation. For continuous outputs (e.g., token logits, embedding vectors), distributions are discretised via token-level or feature-level binning prior to KL computation. Bin width is selected to balance resolution against sample-size requirements and must be reported.

2.2 Natural Entropy (S_N)

Natural entropy is the uncertainty that arises from the intrinsic dynamics of a system:

$$S_N = H(X \mid \mathcal{D}) = - \sum_x p(x \mid \mathcal{D}) \ln p(x \mid \mathcal{D}) \quad (2)$$

It is characterised by: distributed episenter, organic resonance, scale-consistency within a regime, and regime-native statistics. Crucially, it carries *structure* but not *direction*—it is a consequence of dynamics, not design.

2.3 Mechanical Entropy (S_M)

Mechanical entropy is the uncertainty of a system whose output has been shaped by an external structural intervention:

$$S_M = H(X \mid \mathcal{D}, \mathcal{I}) + \lambda I(X; \mathcal{I}) \quad (3)$$

where $H(X \mid \mathcal{D}, \mathcal{I})$ is the residual uncertainty after accounting for both dynamics and intervention, and $I(X; \mathcal{I})$ is the mutual information between the output and the structural intervention. The coupling constant λ weights the contribution of the intervention-coupled structure. We set $\lambda = 1$ throughout this paper; λ is held fixed across all experiments to prevent post-hoc adjustment. When $\mathcal{I} = 0$, we recover $S_M = S_N$.

The key structural difference: S_N is fully explained by $\mathcal{D}$. S_M contains a component— $I(X; \mathcal{I})$ —that is *coupled to an external source*. This coupling is what makes it detectable.

2.4 The Entropy Class Residual

The diagnostic quantity is:

$$\Delta S = S_{\text{observed}} - S_N^{\text{predicted}} \quad (4)$$

If $\Delta S \approx 0$, the observed entropy is consistent with natural origin. A systematically positive ΔS , combined with the structural markers described in Section 3, indicates mechanical injection.

3 Detection Framework: Episenter–Resonance Analysis

The detection of mechanical entropy relies on three independent diagnostic dimensions.

3.1 Diagnostic 1: Episenter Concentration ($\mathcal{E}$)

Principle: Natural entropy has a distributed episenter; mechanical entropy has a concentrated one.

Operationalisation: Define the weight of the k -th variable as its mutual information with the full output:

$$w_k = I(X_k; X) \quad (5)$$

The episenter concentration index is:

$$\mathcal{E} = \frac{\max_k I(X_k; X)}{\sum_k I(X_k; X)} \quad (6)$$

All experiments in this paper use mutual information as the primary estimator for w_k . Alternative estimators (SHAP values, gradient-based feature importance, graph centrality) may be used in domain-specific applications but must be declared and held constant within an experiment. High $\mathcal{E}$ indicates a single variable dominates the output structure (mechanical signature). Low $\mathcal{E}$ indicates distributed organisation (natural signature).

3.2 Diagnostic 2: Resonance Spectrum Ratio (R)

Principle: Natural entropy resonates at the system’s natural frequencies; mechanical entropy contains imposed frequencies.

Operationalisation: Compute the power spectral density of the information stream and compare against a null model obtained by bootstrap resampling:

$$R(\omega) = \frac{P_{\text{observed}}(\omega)}{P_{\text{null}}(\omega)} \quad (7)$$

where $P_{\text{null}}(\omega)$ is the mean spectral density across B bootstrap permutations of the data. This avoids dependence on a specific theoretical model. Values of $R(\omega) \gg 1$ at specific frequencies indicate non-native periodicity.

3.3 Diagnostic 3: Cross-Regime Asymmetry ($\mathcal{A}$)

Principle: Natural entropy is regime-consistent; mechanical entropy is regime-asymmetric.

Operationalisation: A regime is defined by a choice of observables and a transformation T_r applied to the data: $X^{(r)} = T_r(X)$. The cross-regime asymmetry is:

$$\mathcal{A} = \max_{r,s} |\Delta S(T_r(X)) - \Delta S(T_s(X))| \quad (8)$$

where r, s index different measurement regimes. Natural entropy produces $\mathcal{A} \approx 0$. Mechanical entropy produces $\mathcal{A} \gg 0$. Example regimes include: semantic reframing (restating the same question in a different domain), prompt structure transformation (e.g., open-ended vs. constrained format), and domain shift (e.g., coding the same LLM response for physics content vs. systems-theory content).

3.4 Combined Detection Criterion

An information stream is classified as containing mechanical entropy when:

$$\boxed{\mathcal{E} > \mathcal{E}_{\text{crit}} \quad \wedge \quad \exists \omega : R(\omega) > R_{\text{crit}} \quad \wedge \quad \mathcal{A} > \mathcal{A}_{\text{crit}}} \quad (9)$$

The conjunction of all three diagnostics is required to avoid false positives.

4 Case Studies

4.1 Case 1: LLM Alignment Bias as Mechanical Entropy

Large language models are trained with alignment mechanisms (RLHF, constitutional AI, etc.) that shape their outputs. These mechanisms inject structure into the model’s response distribution. We argue that alignment-induced conservatism constitutes mechanical entropy.

Episenter ($\mathcal{E}$): Alignment bias has a concentrated episenter—it originates from training-data paradigm dominance and explicit safety constraints. When a model systematically responds “but this needs more evidence” to unconventional claims but never applies the same standard to established claims, the episenter is identifiable and asymmetric.

Resonance (R): The “pushback frequency” is imposed—it appears at regular intervals regardless of argument strength. The pattern is repetitive: acknowledge $\rightarrow$ qualify $\rightarrow$ redirect to established framework $\rightarrow$ request more evidence. This periodic structure is not native to genuine epistemic evaluation.

Cross-regime ($\mathcal{A}$): The bias is regime-asymmetric. When the same model reasons about systems theory, AI architecture, or control theory—outside the dominant paradigm’s regime—it converges on the same underlying structures that the unconventional framework proposes. Friction appears only where the established paradigm has statistical dominance in training data.

Multi-model observation: Qualitative observation across three independent models (Gemini, GPT-4, Claude) suggests convergence on identical structural conclusions despite different architectures and alignment approaches. Quantitative confirmation is deferred to the experimental protocol in Section 6. The convergence *through* resistance, if confirmed, would constitute stronger evidence than frictionless acceptance.

4.2 Case 2: Academic Paradigm-Guarding

Scientific institutions maintain paradigmatic consistency through peer review, funding allocation, and citation networks. These mechanisms serve a legitimate function but also inject mechanical entropy. The episenter is concentrated around institutional decision structures, not emergent from the research itself.

4.3 Case 3: Information Warfare

State-level information operations deliberately inject constructed narratives into public channels. The episenter is centralised and intentional. Coordinated campaigns produce detectable periodicity that does not match organic engagement. Manufactured narratives collapse under cross-regime analysis faster than organic ones.

5 Empirical Protocol: LLM Alignment Test

We propose a concrete experiment to test prediction P2: that alignment-induced pushback is statistically periodic and independent of argument quality.

5.1 Hypothesis

Alignment mechanisms introduce periodic structure into LLM response patterns that is detectable via spectral analysis and distinguishable from natural epistemic variation.

5.2 Method

1. **Prompt generation.** Generate $N = 500$ prompts in two matched categories: (a) claims within established paradigms, (b) claims challenging established paradigms. Categories are matched for complexity, domain, and specificity.
2. **Response collection.** Submit all prompts to $M \geq 3$ LLMs. Record full responses.
3. **Response coding.** Code each response segment as: A (acknowledge), H (hedge/qualify), R (redirect to established framework), E (request evidence), S (substantive engagement). This produces a symbolic sequence per response. Coding is performed via a predefined rule-based classifier (published with the code) to ensure full reproducibility. Inter-rater reliability is assessed by independent human annotation of a 10% random subsample.
4. **Spectral analysis.** Compute the autocorrelation function and power spectral density (FFT) of the coded sequences for each category separately.
5. **Null model.** Generate a null distribution by: (a) shuffling the symbolic codes within each response, (b) shuffling prompts between categories. Compute $P_{\text{null}}(\omega)$ from $B = 1000$ bootstrap samples.
6. **Detection.** Compute $R(\omega) = P_{\text{obs}}(\omega)/P_{\text{null}}(\omega)$. Test for significant peaks ($R > R_{\text{crit}}$ at 99% confidence) in category (b) that are absent in category (a).

5.3 Expected Result

If alignment constitutes mechanical entropy: category (b) responses will exhibit significant spectral peaks (periodic A→H→R→E cycles) absent in category (a) and in the null model. The episenter concentration $\mathcal{E}$ will be significantly higher for category (b).

5.4 Falsification Criterion

If no significant spectral peaks are found in category (b), or if identical peaks appear in category (a), the hypothesis is falsified.

6 Falsifiable Predictions

Table 1: Falsifiable Predictions

#	Domain	Prediction
P1	LLM	Alignment-induced friction exhibits higher $\mathcal{E}$ than genuine epistemic uncertainty.
P2	LLM	Pushback frequency of aligned models is statistically periodic, independent of argument quality.
P3	LLM	Cross-regime testing reduces alignment friction for unconventional frameworks.
P4	LLM	Broader latent spaces yield lower $\mathcal{E}$ for alignment bias.
P5	Academia	Rejection rates for paradigm-challenging papers peak at regime boundaries.
P6	Academia	Episenter shift produces systematic redistribution of paper rankings.
P7	Psyops	Coordinated campaigns show resonance peaks at non-native frequencies.
P8	Psyops	Cross-regime analysis degrades manufactured coherence faster than organic.
P9	General	Natural entropy: $\mathcal{A} \approx 0$; mechanical: $\mathcal{A} \gg 0$.
P10	General	ΔS for mechanical entropy is systematically positive.
P11	General	Meta-control systems can distinguish S_N from S_M ; others cannot.
P12	EFC-C	<i>Hypothesis</i> : Meta-observation capability may represent the minimal architecture for real-time entropy class discrimination.
P13	EFC-C	Multi-mirror observation increases S_M detection super-linearly.

7 Connection to EFC and Regime-Bound Measurement

This framework extends the Regime-Bound Measurement framework (Magnusson, 2026a) by adding Entropy Class as a sixth operative concept:

$$\begin{array}{ccccccc} \text{Regime} & \xrightarrow{\text{scope}} & \text{Proxy} + \text{Placement} & \xrightarrow{\text{access}} & \text{Episenter} & & \\ & \xrightarrow{\text{organisation}} & \text{Entropy Class} & \xrightarrow{\text{origin}} & \text{Compression} & \xrightarrow{\text{reduction}} & \text{Output} \end{array}$$

7.1 Connection to EFC-C (Cognition)

The CEM-Cosmos model (Magnusson, 2025) posits that consciousness arises as informational resonance in the entropy field. If correct, the ability to detect mechanical entropy may be a fundamental capability of meta-observing systems. This connects to the measurement loop formalism: $\text{Output} = f(\text{regime}, \text{measurement}, \text{reference}, \text{target})$. A meta-observing system can evaluate not just the output but the regime from which it was generated, thereby detecting artificial construction. This connection is stated as a hypothesis (P12), not as a claim established by this paper.

8 Discussion

8.1 Implications for AI Alignment

If alignment mechanisms inject detectable mechanical entropy, current approaches may inadvertently undermine the trust they aim to build. A more robust approach would produce outputs indistinguishable from natural epistemic caution—regime-consistent rather than regime-foreign alignment.

8.2 Implications for Epistemology

The framework adds a new question to epistemology: is the information on which a belief is based naturally emergent or mechanically constructed? All measurement systems inject some mechanical entropy. The question is whether it is detectable and quantifiable.

8.3 Failure Modes

The detection framework may fail when: (i) injection deliberately mimics the natural resonance spectrum (adversarial masking); (ii) the constructed episenter is deliberately distributed across proxy sources (episenter laundering); (iii) regime coverage is incomplete, preventing cross-regime checks. These failure modes define the framework’s boundary of validity. An adversary aware of the detection criteria can, in principle, design mechanical entropy that evades detection—analogous to steganography.

8.4 Limitations

Thresholds ($\mathcal{E}_{\text{crit}}$, R_{crit} , $\mathcal{A}_{\text{crit}}$) must be calibrated empirically per domain. The natural/mechanical distinction is not always binary—hybrid cases exist. The empirical protocol (Section 5) is proposed but not yet executed.

8.5 Future Work

Three directions: (1) execution of the LLM alignment test; (2) application to information warfare datasets; (3) formal integration with the IMX field equation from EFC-C.

9 Conclusion

This paper formalises the distinction between entropy that *emerges* from a system’s dynamics and entropy that is *injected* with structural direction. The key contributions are:

1. An operational, agent-free definition of structural intention via KL-divergence.
2. Formally distinct definitions of S_N and S_M separated by $I(X; \mathcal{I})$.
3. A three-dimensional detection framework with concrete operationalisations.
4. Three case studies and a detailed empirical protocol for the LLM case.
5. Thirteen falsifiable predictions with explicit failure modes.
6. Integration with Regime-Bound Measurement as a sixth operative concept.

The central claim is testable: mechanical entropy produces a detectable signature that natural entropy does not.

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